Resampling Techniques

Problem 6: 2D Spatial Smoothing

In image processing, an image is a 2D signal $I(x, y)$. When an image is corrupted by noise, we can estimate the true pixel value using **2D Nadaraya-Watson Regression**:

$$\hat{I}(x, y) = \frac{\sum_{i=1}^{n} K_h(d_i) \cdot y_i}{\sum_{i=1}^{n} K_h(d_i)}$$

where:

- (x, y) is the coordinate of the pixel we are reconstructing.
- $d_i = \sqrt{(x - x_i)^2 + (y - y_i)^2}$ is the Euclidean distance to the i-th neighbor.
- y_i is the observed (noisy) intensity of the neighbor pixel.

Choosing the Kernel $K_h(u)$

Students must implement these three "spatial filters":

1. **Boxcar (Uniform):** Simple average of the neighborhood.

$$K(u) = \mathbb{I}(u \leq 1)$$

2. **Gaussian:** Smooth weighted average (Gaussian Blur).

$$K(u) = \exp(-0.5u^2)$$

3. **Epanechnikov:** Parabolic weighting (Optimal MSE).

$$K(u) = \max(0, 0.75(1 - u^2))$$

Note: $u = dist/h$. Small h keeps the image sharp but noisy; large h removes noise but blurs the image.

Implementation: Image as Data

To make this computationally feasible for the lab:

1. Take a small grayscale image (e.g., 64×64 pixels).
2. Add Gaussian noise to create a "Noisy Image."
3. **The Task:** For every pixel (x, y), use its surrounding neighbors within a radius h to calculate the "smoothed" value.

Note: Unlike 1D signals, x and y now represent spatial grid coordinates (row/column indices).

Bootstrap for Image Quality

How do we know which kernel preserves edges better without knowing the "true" clean image? We use **Bootstrap** to check stability.

The Bootstrap Procedure:

1. Treat the pixels in a noisy patch as your population.
2. Generate $B = 50$ bootstrap samples of pixels from that patch.
3. Apply 2D Kernel Regression using all 3 kernels.
4. Calculate the **MSE** between the reconstructed patch and the original.
5. Compare the distribution of errors via Boxplots.

Tasks 6.1

1. Load a grayscale image of your choice and downsample it to 64×64 (cropping or scaling). You can use the "PIL" library for image loading, conversion to grayscale, etc or you can generate your own image.
2. Add random noise: $I_{\text{noisy}} = I_{\text{true}} + \sigma \mathcal{N}(0, 1)$. Take sigma to be 5% of maximuma pixel value.
3. Implement 2D NW-Regression for all 3 kernels.
4. Use Bootstrap to estimate the MSE distribution for both filters.
5. **Deliverable:** Show the Original, Noisy, and Denoised images side-by-side.

Task 6.2: Bandwidth Sweep (h-Optimization)

How does the choice of h affect the balance between noise reduction and detail preservation?

1. Select a range of bandwidth values: $h \in \{1.0, 2.0, \cdots, 6.0\}$.
2. For each h, perform the 2D Kernel Regression using all 3 kernels..
3. Calculate two metrics for each denoised image:
    - **MSE (Mean Squared Error):** Measures absolute pixel difference (Lower is better).
    - **SSIM (Structural Similarity):** Measures perceived quality based on structure, luminance, and contrast (Range: 0–1, 1 is perfect).

Implementation Note: Students can find these functions in the `scikit-image` library:

- `from skimage.metrics import mean_squared_error as mse`
- `from skimage.metrics import structural_similarity as ssim`

Deliverable: The Dual-Axis Performance Plot

Produce a chart showing the relationship between h and image quality. Use $h = 2, 3, 4, 5$ for each kernel.

Expected Visualization:

- **Primary Y-axis (Left):** MSE (should show a U-shaped curve).
- **Secondary Y-axis (Right):** SSIM (should show an inverted U-shape).
- **X-axis:** Bandwidth value (h).

Note: The "Optimal" h for MSE may not be the same as the "Optimal" h for SSIM. Discuss why in your report.

Interpretation Questions

- Which kernel produces a "blocky" or "blurred" artifact? Why?
- Why does the Gaussian or Epanechnikov kernel usually result in a more visually pleasing image than the Boxcar?
- Look at the edges. Why do both kernels struggle to keep edges sharp while removing noise?
- **The *h*-Sweep:** Looking at your dual-axis plot, does the h that minimizes MSE also maximize the SSIM? Why might these "optimal" points differ?
- **Metric Sensitivity:** Which metric (MSE or SSIM) is more sensitive to the loss of fine details (like edges) as h increases?
- **Bias-Variance Tradeoff:** In the context of denoising, what does a very low h represent in terms of Bias and Variance? What about a very high h?
- **Real-world Application:** If you were denoising a medical X-ray where structural integrity is life-critical, which metric would you prioritize and why?